

Polar Equations and Graphs

ANSWER KEY



Recognizing polar graphs

1 Identify the graph of each polar equation:

- a** $r = 3$ Circle of radius 3 centered at the pole. **b** $\theta = \frac{\pi}{6}$ Line through the pole at angle $\frac{\pi}{6}$.
c $r = 2 + 2\cos \theta$ Cardioid (heart-shaped, symmetric about the polar axis). **d** $r = 4\sin 3\theta$ Rose curve with 3 petals.

2 For the limaçon $r = 1 + 2\cos \theta$:

- a** Explain how you know it has an inner loop. In $r = a + b\cos \theta$, $|a| < |b|$ (here $1 < 2$) produces an inner loop.
b Find the values of θ in $[0, 2\pi)$ at which the curve passes through the pole. $1 + 2\cos \theta = 0$ gives $\cos \theta = -\frac{1}{2}$: $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$.

3 For the rose $r = 5\sin 2\theta$:

- a** State the number of petals. $n = 2$ is even, so $2n = 4$ petals.
b State the maximum value of $|r|$ and the smallest positive θ at which it occurs. Maximum $|r| = 5$, first at $2\theta = \frac{\pi}{2}$, i.e. $\theta = \frac{\pi}{4}$.

4 Determine whether the graph of $r = 2 + \sin \theta$ passes through the pole, and state the type of limaçon. $2 + \sin \theta \geq 1 > 0$ for all θ , so it never passes through the pole. Since $|a| > |b|$ ($2 > 1$), it is a dimpled limaçon with no inner loop.

Symmetry tests

5 Test $r = 3\cos 2\theta$ for symmetry about the polar axis (replace θ with $-\theta$) and about the pole (replace r with $-r$).

Polar-axis: $\cos(2(-\theta)) = \cos(2\theta)$, unchanged — symmetric about the polar axis. Pole: replacing $r \rightarrow -r$ gives $-r = 3\cos 2\theta$, not equivalent to the original in general — the standard test is inconclusive for pole symmetry, though the curve (a 4-petal rose) is in fact symmetric about the pole by direct graphing.

6 Test $r = 2 + 2\sin \theta$ for symmetry about the line $\theta = \frac{\pi}{2}$ (replace θ with $\pi - \theta$).

$\sin(\pi - \theta) = \sin \theta$, so the equation is unchanged: symmetric about $\theta = \frac{\pi}{2}$ (the vertical axis).

Graphing from a table of values

- 7 For the cardioid $r = 1 + \cos \theta$, complete the table of r -values: $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$.
- a $r(0) = 2$ b $r(\frac{\pi}{2}) = 1$ c $r(\pi) = 0$ d $r(\frac{3\pi}{2}) = 1$
- 8 For the circle $r = 4\sin \theta$, find the value(s) of θ in $[0, \pi)$ where r reaches its maximum, and state that maximum value.
- Maximum of $\sin \theta$ is 1 at $\theta = \frac{\pi}{2}$, giving $r = 4$.
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Rose curves in depth

- 9 For the rose $r = 3\cos 5\theta$, state the number of petals and the length of each petal (maximum $|r|$).
- $n = 5$ is odd, so there are $n = 5$ petals, each of length $|r|_{\max} = 3$.
- 10 Explain, using the general rule for rose curves $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$, why n odd gives n petals but n even gives $2n$ petals.
- When n is odd, tracing θ from 0 to 2π retraces each petal a second time (due to negative r values overlapping earlier petals), so only n distinct petals appear. When n is even, no such overlap occurs, and all $2n$ petals from the full $[0, 2\pi)$ sweep are distinct.
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Identifying limaçon subtypes

- 11 Classify each limaçon $r = a + b\cos \theta$ as having an inner loop, being dimpled, or being convex (roughly oval, no dimple), based on the ratio $\frac{a}{b}$:
- a $r = 1 + 3\cos \theta$ $\frac{a}{b} = \frac{1}{3} < 1$: inner loop.
- b $r = 3 + 2\cos \theta$ $1 < \frac{a}{b} = \frac{3}{2} < 2$: dimpled.
- c $r = 4 + \cos \theta$ $\frac{a}{b} = 4 > 2$: convex (no dimple).
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Finding maximum and minimum r-values

- 12 For the limaçon $r = 3 + 2\sin \theta$, find the maximum and minimum values of r , and the angles at which they occur.
- Maximum $r = 5$ at $\sin \theta = 1$ ($\theta = \frac{\pi}{2}$). Minimum $r = 1$ at $\sin \theta = -1$ ($\theta = \frac{3\pi}{2}$).
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Comparing sine- and cosine-based polar curves

- 13 Explain how the graph of $r = 3 + 3\sin \theta$ compares to $r = 3 + 3\cos \theta$ (same cardioid shape, different orientation).

Both are cardioids of the same size and shape. The cosine version $r = 3 + 3\cos \theta$ is symmetric about the polar axis (horizontal), pointing right, while the sine version $r = 3 + 3\sin \theta$ is symmetric about $\theta = \frac{\pi}{2}$ (vertical), pointing up — a 90° rotation of one relative to the other.

Sketching from key features

- 14 Without fully graphing, list the key features you would compute first to sketch $r = 2 - 4\cos \theta$ (max/min r , pole crossings, symmetry).

Type: limaçon with inner loop ($|a| = 2 < |b| = 4$). Symmetry: about the polar axis (cosine-based). Max $|r| = 6$ at $\theta = \pi$; min value $r = -2$ at $\theta = 0$. Pole crossings: $2 - 4\cos \theta = 0 \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}, \frac{5\pi}{3}$ (marking where the inner loop begins/ends).

A capstone graphing problem

- 15 For the rose $r = 4\cos(3\theta)$, find: (a) the number of petals, (b) the maximum value of r , (c) the value(s) of $\theta \in [0, \pi)$ where the first petal reaches its maximum length, and (d) the angles where the curve passes through the pole in $[0, \pi)$.

(a) $n = 3$ is odd: 3 petals. (b) Maximum $|r| = 4$. (c) $\cos(3\theta) = 1 \Rightarrow 3\theta = 0 \Rightarrow \theta = 0$ (first petal's tip). (d) $\cos(3\theta) = 0 \Rightarrow 3\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \Rightarrow \theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$.

Distinguishing curve families by equation form alone

- 16 Without graphing, state whether each polar equation is a circle, a line, a rose, or a limaçon, based purely on its algebraic form: $r = 7$, $r = 5\sin(4\theta)$, $\theta = \frac{2\pi}{5}$, $r = 2 - 2\cos \theta$.
 $r = 7$: circle (constant r). $r = 5\sin(4\theta)$: rose ($n = 4$ even, 8 petals). $\theta = \frac{2\pi}{5}$: line through the pole (constant θ). $r = 2 - 2\cos \theta$: cardioid (a special limaçon, coefficient equals constant).